

1. Launch one aws ec2 and install jenkins.

```
ubuntu@ip-172-31-24-128: ~  
91738@IDEAPAD MINGW64 ~  
$ cd ~/Downloads  
91738@IDEAPAD MINGW64 ~/Downloads  
$ chmod 400 jenkins-key.pem  
91738@IDEAPAD MINGW64 ~/Downloads  
$ ssh -i jenkins-key.pem ubuntu@34.233.126.119  
ssh: connect to host 34.233.126.119 port 22: Connection timed out  
91738@IDEAPAD MINGW64 ~/Downloads  
$ ^C  
91738@IDEAPAD MINGW64 ~/Downloads  
$ ssh -i jenkins-key.pem ubuntu@34.233.126.119  
ssh: connect to host 34.233.126.119 port 22: Connection timed out  
91738@IDEAPAD MINGW64 ~/Downloads  
$ ssh -i jenkins-key.pem ubuntu@34.233.126.119  
The authenticity of host '34.233.126.119 (34.233.126.119)' can't be established.  
ED25519 key fingerprint is SHA256:KqNsVpN4cxkZT/gsx8BnkHofbQz14tZxfBzsky7g.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '34.233.126.119' (ED25519) to the list of known hosts  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Mon Aug 10 10:30:49 UTC 2026  
  
System load: 0.04          Temperature: ~273.1 C  
Usage of /: 30.5% of 6.61GB Processes:      116  
Memory usage: 25%         Users logged in: 0  
Swap usage: 0%            IPv4 address for ens5: 172.31.24.128  
  
Expanded Security Maintenance for Applications is not enabled.  
0 updates can be applied immediately.  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@ip-172-31-24-128:~$ java -version  
openjdk version "17.0.19" 2026-04-21  
OpenJDK Runtime Environment (build 17.0.19+10-1-26.04.2-Ubuntu)  
OpenJDK 64-Bit Server VM (build 17.0.19+10-1-26.04.2-Ubuntu, mixed mode, sharing)  
ubuntu@ip-172-31-24-128:~$ sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \  
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key  
--2026-08-10 10:38:13-- https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key  
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.38.133, 2a04:4e42:95::645  
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.38.133|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 1680 (1.6K) [application/octet-stream]  
Saving to: '/etc/apt/keyrings/jenkins-keyring.asc'  
  
/etc/apt/keyrings/jenkins-keyring.asc 100%[=====]  
  
2026-08-10 10:38:13 (25.0 MB/s) - '/etc/apt/keyrings/jenkins-keyring.asc' saved [1680/1680]  
  
ubuntu@ip-172-31-24-128:~$ echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] https://pkg.jenkins  
/etc/apt/sources.list.d/jenkins.list" > /dev/null  
ubuntu@ip-172-31-24-128:~$ sudo apt update  
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease  
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease  
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease  
Hit:4 http://security.ubuntu.com/ubuntu InRelease  
Ign:5 https://pkg.jenkins.io/debian-stable binary/ InRelease  
Get:6 https://pkg.jenkins.io/debian-stable binary/ Release [2044 B]  
Get:7 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]  
Get:8 https://pkg.jenkins.io/debian-stable binary/ Packages [31.4 kB]
```

```
ubuntu@ip-172-31-24-128:~$ sudo apt install fontconfig openjdk-21-jre -y
fontconfig is already the newest version (2.17.1-3ubuntu1).
The following package was automatically installed and is no longer required:
  polinate
Use 'sudo apt autoremove' to remove it.

Installing:
  openjdk-21-jre

Installing dependencies:
  ibus-gtk3 libcolord2 libepoxy0 libgtk-3-0t64 libgtk-3-bin libgtk-3-common libwayland-cursor0 libwayland-egl1 openjdk-21-jre-headless

Suggested packages:
  colord gvfs libnss-mdns fonts-ipafont-gothic fonts-ipafont-mincho fonts-wqy-microhei fonts-wqy-zenhei fonts-indic

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 5
  Download size: 52.0 MB
  Space needed: 221 MB / 3713 MB available

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu/resolute/main amd64v3 ibus-gtk3 amd64 1.5.34-rc2-1 [19.9 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu/resolute/main amd64v3 libcolord2 amd64 1.4.8-3 [151 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu/resolute/main amd64v3 libepoxy0 amd64 1.5.10-2build1 [226 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu/resolute/main amd64v3 libwayland-cursor0 amd64 1.24.0-2 [10.8 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu/resolute/main amd64v3 libwayland-egl1 amd64 1.24.0-2 [6264 B]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu/resolute/main amd64v3 libgtk-3-common all 3.24.52-0ubuntu1 [1424 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu/resolute/main amd64v3 libgtk-3-0t64 amd64 3.24.52-0ubuntu1 [3085 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu/resolute/main amd64v3 libgtk-3-bin amd64 3.24.52-0ubuntu1 [74.3 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu/resolute-updates/universe amd64v3 openjdk-21-jre-headless amd64 21.0.11+10-1-26.04.2 [46.8 MB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu/resolute-updates/universe amd64v3 openjdk-21-jre amd64 21.0.11+10-1-26.04.2 [227 kB]
Fetched 52.0 MB in 1s (93.9 MB/s)
Selecting previously unselected package ibus-gtk3:amd64.
(Reading database ... 132337 files and directories currently installed.)
Preparing to unpack .../0-ibus-gtk3_1.5.34-rc2-1_amd64v3.deb ...
Unpacking ibus-gtk3:amd64 (1.5.34-rc2-1) ...
Selecting previously unselected package libcolord2:amd64.
Preparing to unpack .../1-libcolord2_1.4.8-3_amd64v3.deb ...
Unpacking libcolord2:amd64 (1.4.8-3) ...
Selecting previously unselected package libepoxy0:amd64.
Preparing to unpack .../2-libepoxy0_1.5.10-2build1_amd64v3.deb ...
Unpacking libepoxy0:amd64 (1.5.10-2build1) ...
Selecting previously unselected package libwayland-cursor0:amd64.
Preparing to unpack .../3-libwayland-cursor0_1.24.0-2_amd64v3.deb ...
Unpacking libwayland-cursor0:amd64 (1.24.0-2) ...
Selecting previously unselected package libwayland-egl1:amd64.
Preparing to unpack .../4-libwayland-egl1_1.24.0-2_amd64v3.deb ...
Unpacking libwayland-egl1:amd64 (1.24.0-2) ...
```

```
ubuntu@ip-172-31-24-128:~$ sudo apt install jenkins -y
The following package was automatically installed and is no longer required:
  polinate
Use 'sudo apt autoremove' to remove it.

Installing:
  jenkins

Installing dependencies:
  net-tools

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 5
  Download size: 101 MB
  Space needed: 102 MB / 3815 MB available

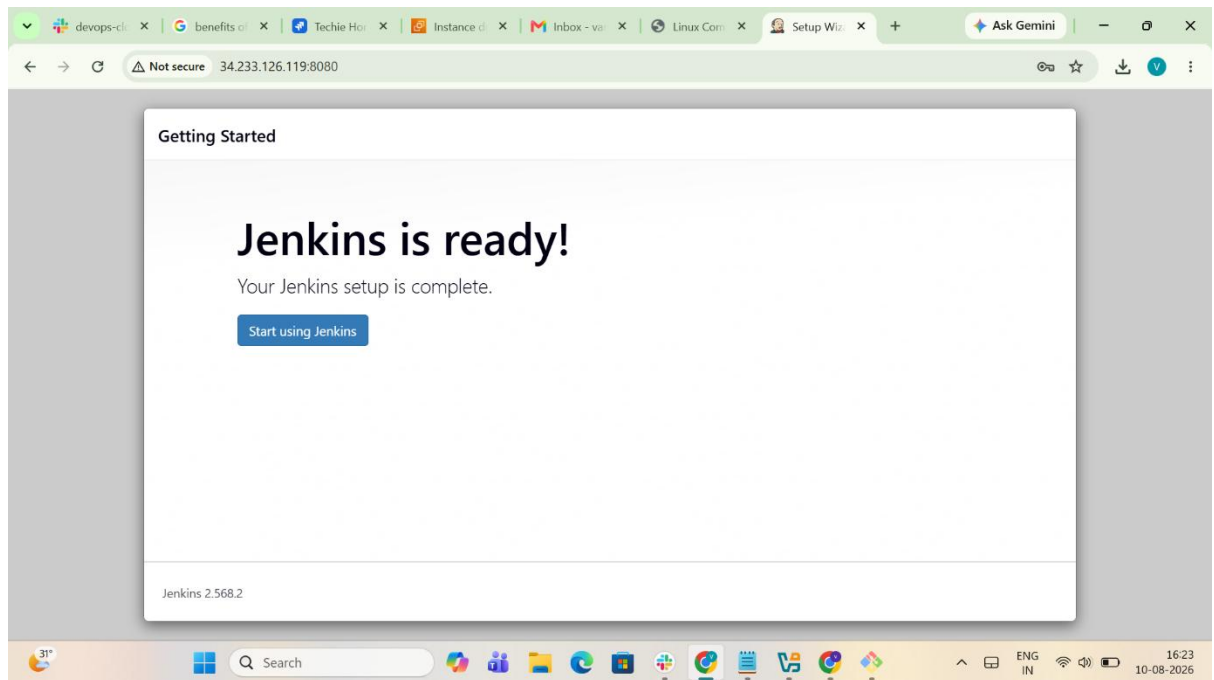
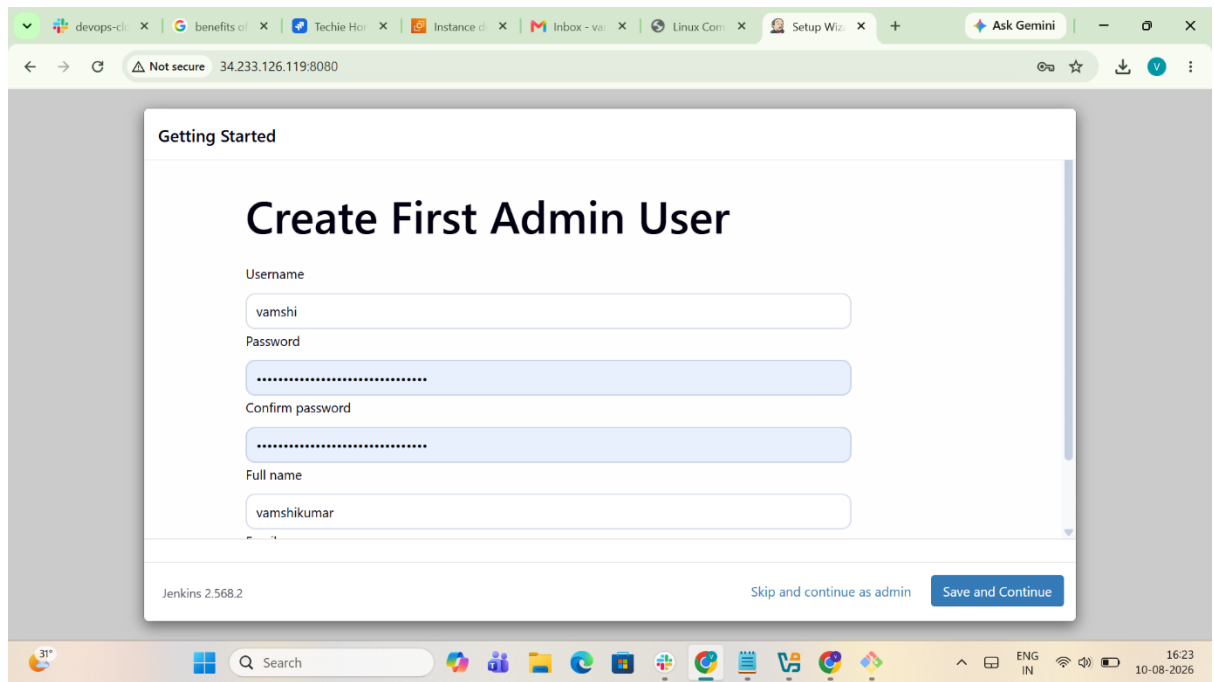
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu/resolute/main amd64v3 net-tools amd64 2.10-2ubuntu1 [152 kB]
Get:2 https://pkg.jenkins.io/debian-stable binary/ jenkins 2.568.2 [101 MB]
Fetched 101 MB in 1s (73.1 MB/s)
```

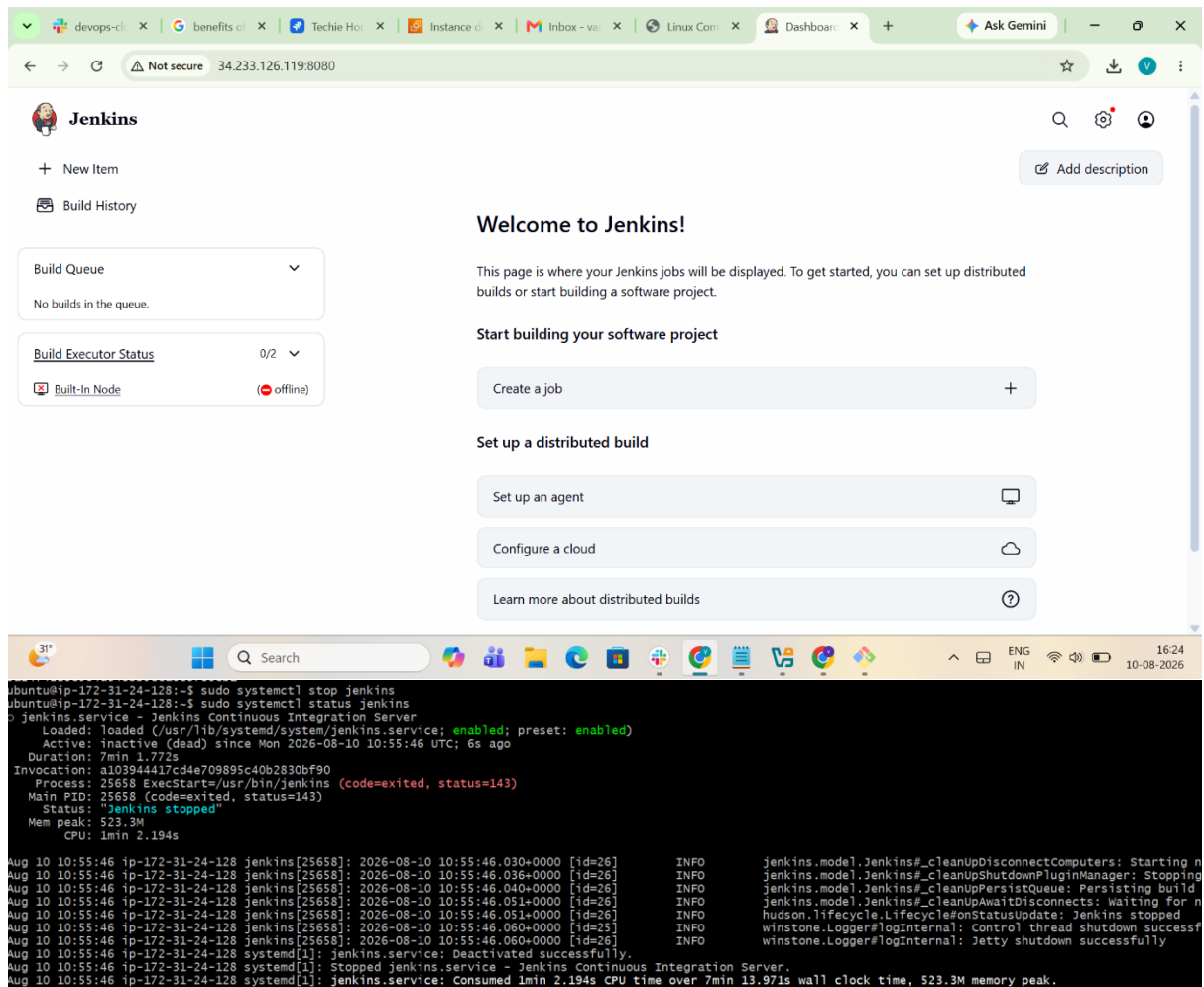
```
ubuntu@ip-172-31-24-128:~$
No containers need to be restarted.

User sessions running outdated binaries:
  ubuntu @ session #1: sshd-session[1156,1273]
  ubuntu @ user manager: (sd-pam)[1163]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-24-128:~$ java -version
openjdk version "21.0.11" 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-26.04.2-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-24-128:~$ readlink -f $(which java)
/usr/lib/jvm/java-21-openjdk-amd64/bin/java
ubuntu@ip-172-31-24-128:~$ sudo systemctl start jenkins
ubuntu@ip-172-31-24-128:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-10 10:48:44 UTC; 14s ago
     Invocation: a103944417cd4e709895c40b2830bf90
       Main PID: 25658 (java)
         Tasks: 41 (limit: 627)
        Memory: 314.9M (peak: 319.3M)
           CPU: 20.894s
      CGroup: /system.slice/jenkins.service
              └─25658 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Aug 10 10:48:39 ip-172-31-24-128 jenkins[25658]: [LF]>
Aug 10 10:48:39 ip-172-31-24-128 jenkins[25658]: [LF]>
Aug 10 10:48:39 ip-172-31-24-128 jenkins[25658]: [LF]>
Aug 10 10:48:39 ip-172-31-24-128 jenkins[25658]: [LF]>
Aug 10 10:48:44 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:48:44.126+0000 [id=34] INFO jenkins.InitReactorRunner$1:onAttained: Completed initialization
Aug 10 10:48:44 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:48:44.171+0000 [id=24] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
Aug 10 10:48:44 ip-172-31-24-128 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Aug 10 10:48:44 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:48:44.360+0000 [id=50] INFO hudson.DownloadService$Downloadable#load: Obtained the updated data f
Aug 10 10:48:44 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:48:44.361+0000 [id=50] INFO hudson.util.Retrier#start: Performed the action check updates serv
Aug 10 10:48:49 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:48:49.250+0000 [id=69] WARNING h.n.DiskSpaceMonitorDescriptor#markNodeOfflineOnline: Making
lines 1-21/21 (END)
[LF] Stopped
ubuntu@ip-172-31-24-128:~$ sudo ss -lntlp | grep 8080
LISTEN 0 50 *:8080 *:* users:([ "java",pid=25658,fd=9])
ubuntu@ip-172-31-24-128:~$ cat /var/lib/jenkins/secrets/initialAdminPassword
cat: /var/lib/jenkins/secrets/initialAdminPassword: Permission denied
ubuntu@ip-172-31-24-128:~$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
9a24f4225e564d348bb5ccae3f83eb1
ubuntu@ip-172-31-24-128:~$ sudo systemctl stop jenkins
ubuntu@ip-172-31-24-128:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
```





Purpose

Launch an AWS EC2 instance and install Jenkins.

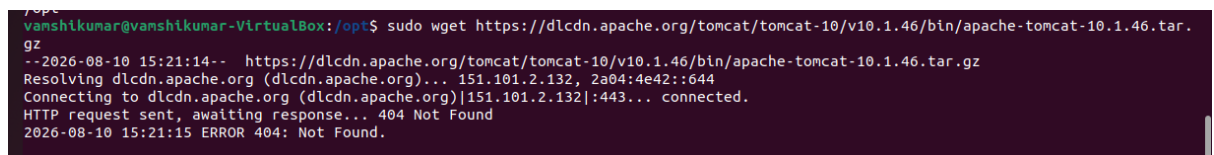
Command Execution & Verification

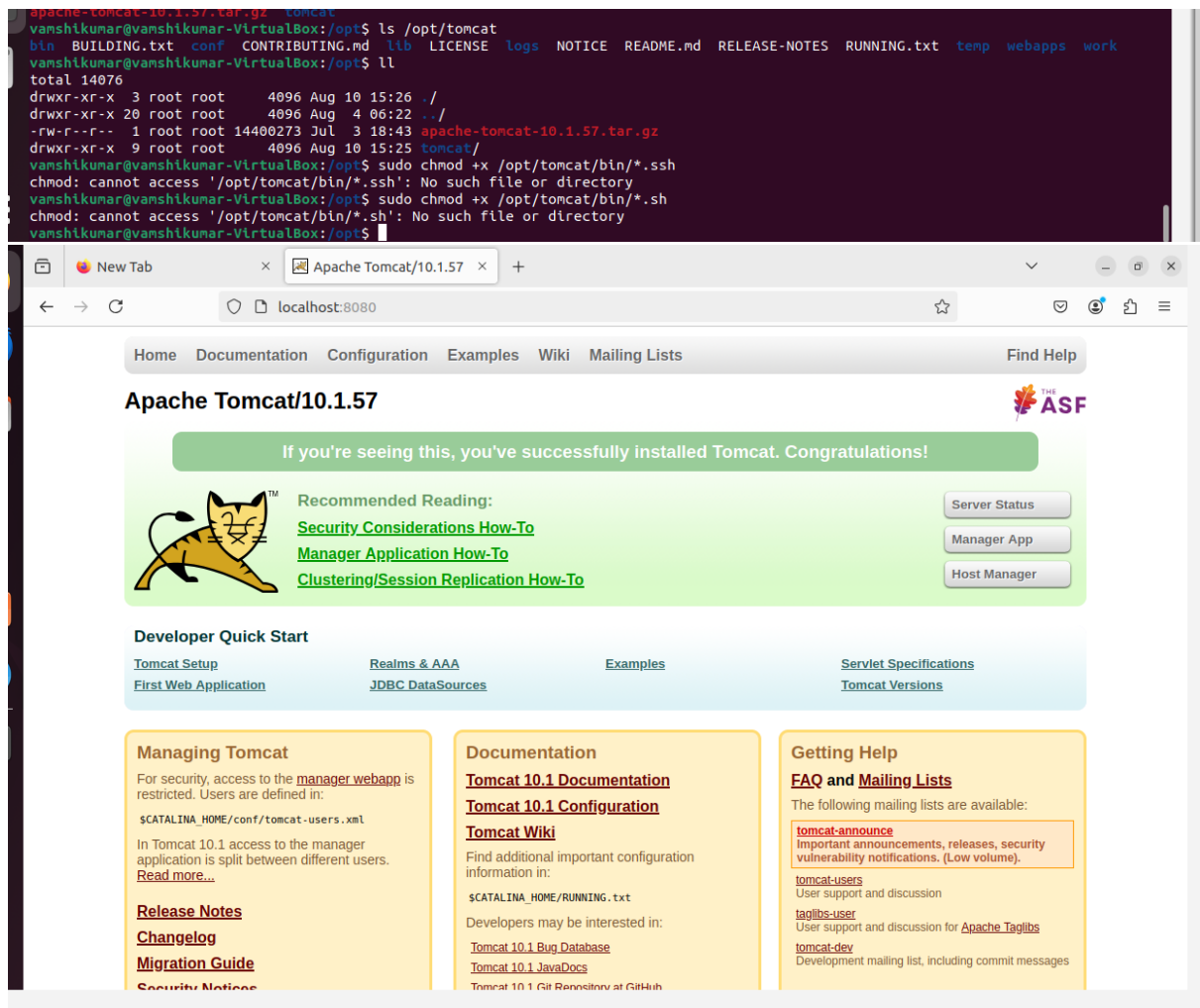
- Launched an Ubuntu EC2 instance and connected using SSH.
- Installed Java 21 and Jenkins.
- Started Jenkins using systemctl.
- Verified Jenkins service status successfully.

Conclusion

Successfully launched an EC2 instance and installed Jenkins.

2. Setup Apache Tomcat on Ubuntu Virtual Machine





Purpose

Setting up an application server like Apache Tomcat allows Java-based web applications and servlets to run and serve dynamic content over HTTP.

Command Execution and Verification

The command `sudo systemctl status tomcat` checks the status of the Tomcat service, returning active (running) with process logs showing initializing ProtocolHandler ["http-nio-8080"] and Server startup in [2251] milliseconds. This verifies that Apache Tomcat is successfully initialized and running on port 8080.

Conclusion

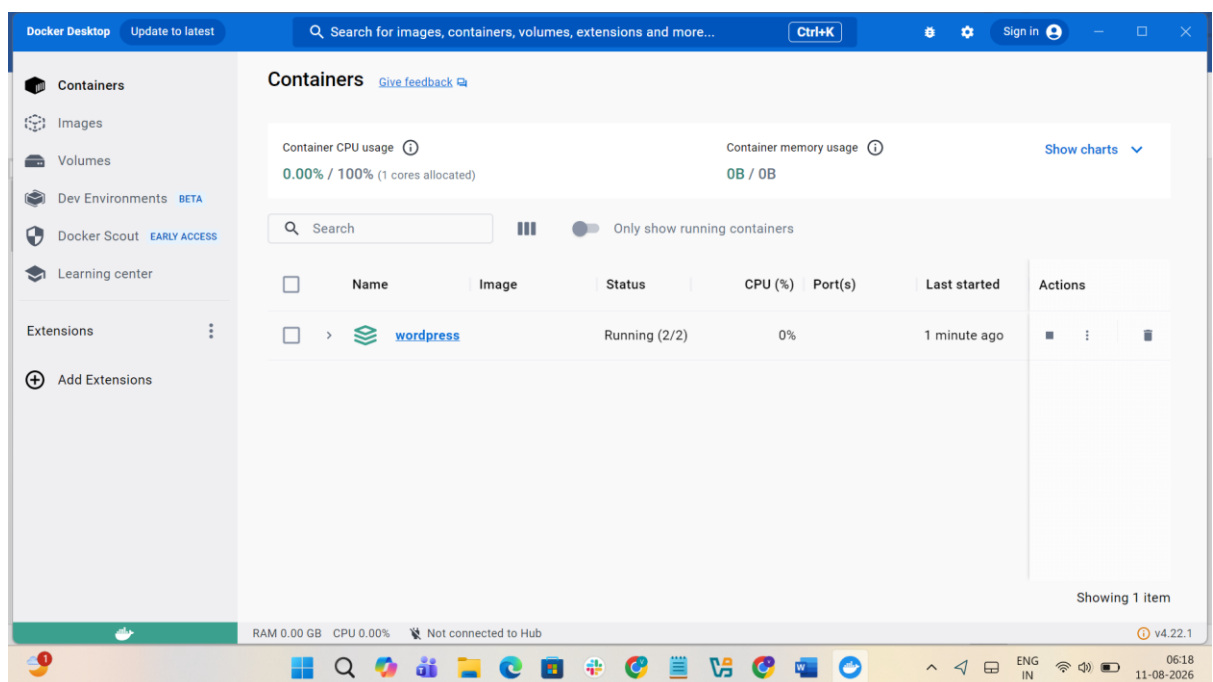
Deploying an application server requires verifying the service state via `systemctl` to confirm that the Java runtime environment and web application directories are deployed and listening on the designated port.

3. Setup Docker Desktop on your local windows.

```
MINGW64:/c/Users/91738

91738@IDEAPAD MINGW64 ~
$ docker --version
Docker version 24.0.5, build ced0996

91738@IDEAPAD MINGW64 ~
$ |
```



Purpose

Set up Docker Desktop on a local Windows machine.

Command Execution & Verification

- Downloaded and installed Docker Desktop.
- Enabled required Windows virtualization features.
- Started Docker Desktop successfully.
- Verified the installation using `docker --version`.

Conclusion

Successfully installed and configured Docker Desktop on Windows.